

THM_ROOMS

INTRODUCTORY NETWORKING

Core Networking Concepts Every Junior Pentester Should Understand

Prepared by

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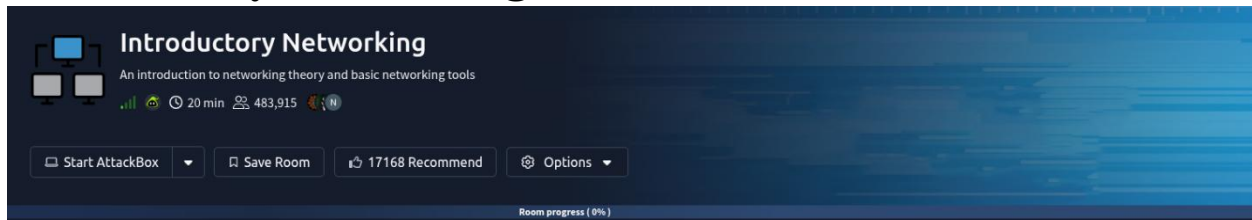
Overview

This document demonstrates my understanding of networking fundamentals from a security perspective and summarizes key networking concepts and tools covered in the *Introductory Networking* room.

The focus is on **understanding**, **technical reasoning**, and **ethical learning**, rather than reproducing lab content or solution

Full write-up: <https://medium.com/@ahmed.a.eldkrory/introductory-networking-thm-room-62866ae1aca7>

Introductory Networking



The aim of this room is to provide a beginner's introduction to the basic principles of networking. Networking is a massive topic, so this really will just be a brief overview; however, it will hopefully give you some foundational knowledge of the topic, which you can build upon for yourself.

The topics that we're going to cover in this room are:

- The OSI Model
- The TCP/IP Model
- How these models look in practice
- An introduction to basic networking tools

Link : <https://tryhackme.com/room/introtonetworking>

Why Networking Matters in Cybersecurity

Networking is not just a theoretical topic — it's the foundation of how systems communicate, how services are exposed, and how attackers and defenders interact.

For penetration testers, a strong networking background enables:

- Accurate reconnaissance and scoping



- Better interpretation of tool output
- Clear technical communication during assessments

Understanding networking early saves time, reduces mistakes, and builds confidence when moving into advanced security topics.

Networking Models: Thinking in Layers

OSI Model – A Diagnostic Framework

The OSI model breaks communication into logical layers, each with a specific role. Rather than memorizing the layers, security professionals use this model to:

- Pinpoint where issues occur
- Understand where attacks are executed
- Communicate findings using a shared technical language

It's a thinking tool — not just a diagram.

There are many mnemonics floating around to help you learn the layers of the OSI model -- search around until you find one that you like.

I personally favour: **Anxious Pale Shakespeare Treated Nervous Drunks Patiently**

<u>OSI:</u>
Application
Presentation
Session
Transport
Network
Data Link
Physical

TCP/IP Model – How the Internet Actually Works

The TCP/IP model reflects real-world implementation. It simplifies networking into practical layers used by modern systems and protocols. Knowing both models helps bridge the gap between theory and hands-on security work.

OSI	TCP/IP
Application	Application
Presentation	
Session	
Transport	Transport
Network	Internet
Data Link	Network Interface
Physical	

Packet Encapsulation: How Data Travels

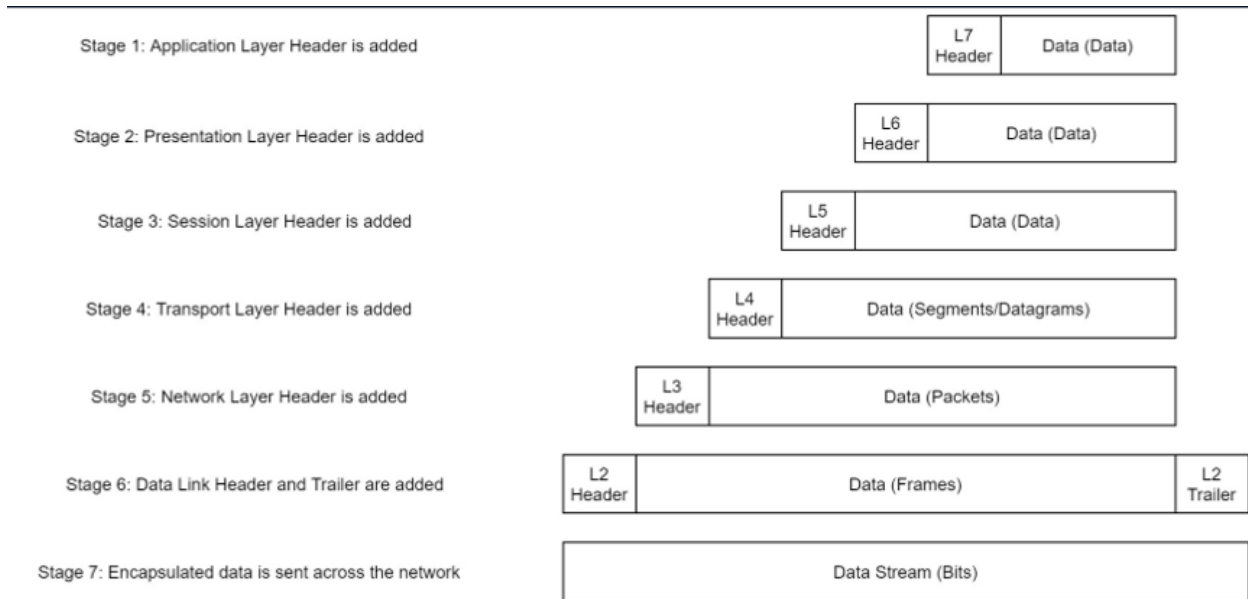
Encapsulation describes how data is wrapped with protocol-specific information as it moves through the network stack.

From a security perspective, this process explains:



- Why packet inspection is powerful
- How metadata is exposed at different layers
- Where monitoring and filtering mechanisms operate

Understanding encapsulation makes traffic analysis far more intuitive.



Essential Networking Tools (High-Level View)

Ping

Used to test basic connectivity and reachability between systems.
Often the first step in validating whether a host is accessible.

Traceroute

Shows the path traffic takes to reach a destination by identifying intermediate network hops.
Useful for understanding routing behavior and network structure.

WHOIS

Retrieves registration and ownership information for domains and IP ranges.
Provides valuable context during reconnaissance and scope validation.



Dig

Queries DNS records directly to understand how domains resolve and are configured.
A key tool for DNS analysis and troubleshooting.

Pentester Mindset: How These Fit Together

These concepts and tools are primarily used during the **reconnaissance phase** of a penetration test.

The goal is to:

- Observe before interacting
- Understand the environment before testing it
- Build a mental map of the network

Good reconnaissance is about **clarity**, not noise.

Common Challenges & Key Insights

Early learners often:

- Focus too much on tools instead of interpretation
- Expect immediate vulnerabilities instead of context
- Underestimate the value of fundamentals

The main takeaway is simple:

Strong fundamentals scale. Weak fundamentals break under pressure.

Key Takeaways

- Networking models improve technical reasoning
 - Encapsulation explains packet-level visibility
 - Simple tools provide powerful insights when used correctly
 - Ethical, passive reconnaissance is a core security skill
 - Fundamentals are a long-term investment in your career
-

Real-World Relevance

In professional security work, these skills support:



- Network reconnaissance and validation
- Incident response investigations
- Security assessments and audits
- Clear reporting and stakeholder communication

Every advanced technique builds on these basics.

Task 1 – Introduction to Networking Concepts

Context

This section introduces the purpose of networking in modern systems and why it is a core skill for cybersecurity professionals.

What This Task Teaches

- Why networking knowledge is essential for penetration testing
- How communication between systems forms the attack surface
- The importance of fundamentals before advanced techniques

Visual Reference

Task 1 Introduction

The aim of this room is to provide a beginner's introduction to the basic principles of networking. Networking is a *massive* topic, so this really will just be a brief overview; however, it will hopefully give you some foundational knowledge of the topic, which you can build upon for yourself.

The topics that we're going to cover in this room are:

- The OSI Model
- The TCP/IP Model
- How these models look in practice
- An introduction to basic networking tools

Answer the questions below

Let's get started!

No answer needed

Check



Task 2 – OSI Model

Context

The OSI model is presented as a structured way to understand how data flows between systems.

Security Perspective

From a penetration tester's viewpoint, the OSI model helps:

- Identify where an issue or weakness exists
- Classify attacks by layer
- Communicate findings clearly in reports

Task

Task 2 ○ The OSI Model: An Overview

The OSI (Open Systems Interconnection) Model is a standardised model which we use to demonstrate the theory behind computer networking. In practice, it's actually the more compact TCP/IP model that real-world networking is based off; however the OSI model, in many ways, is easier to get an initial understanding from.

The OSI model consists of seven layers:

OSI:
Application
Presentation
Session
Transport
Network
Data Link
Physical

There are many mnemonics floating around to help you learn the layers of the OSI model -- search around until you find one that you like.

I personally favour: **A**nxious **P**ale **S**hakespeare **T**reated **N**ervous **D**runks **P**atiently

Q&A

1. Choosing TCP or UDP

Question Concept: Which layer decides between connection-oriented (TCP) and connectionless (UDP) transport?

Answer: Transport Layer (Layer 4)

Explanation: The transport layer is responsible for reliable or fast delivery. TCP provides



reliability, flow control, and ordering, while UDP is faster and connectionless, suitable for real-time applications.

2. Checking Data Integrity

Question Concept: Which layer ensures that received data is not corrupted?

Answer: Data Link Layer (Layer 2)

Explanation: This layer adds error detection codes (like CRC) to detect transmission errors and ensure frames are received correctly.

3. Preparing Data for Transmission

Question Concept: At which layer is data formatted into a structured frame for sending?

Answer: Data Link Layer (Layer 2)

Explanation: Data is packaged into frames with headers/trailers, ready for transmission over physical media.

4. Transmitting and Receiving Data

Question Concept: Which layer handles the physical sending and receiving of bits?

Answer: Physical Layer (Layer 1)

Explanation: This layer converts frames into electrical, optical, or radio signals for actual transmission across the network.

5. Encrypting or Compressing Data

Question Concept: Which layer transforms data into standardized format via encryption or compression?

Answer: Presentation Layer (Layer 6)

Explanation: The presentation layer prepares data for application consumption, handling encryption, compression, and formatting.

6. Tracking Communications



Question Concept: Which layer manages communication sessions between hosts?

Answer: Session Layer (Layer 5)

Explanation: It establishes, maintains, and terminates sessions, ensuring that communications are coordinated and stateful.

7. Accepting Requests from Applications

Question Concept: Which layer interfaces directly with user applications?

Answer: Application Layer (Layer 7)

Explanation: Provides services to applications, enabling programs to send/receive data over the network.

8. Handling Logical Addressing

Question Concept: Which layer is responsible for logical addressing (IP addressing)?

Answer: Network Layer (Layer 3)

Explanation: Routes data packets using logical addresses, determining the best path from source to destination.

9. Data Units in TCP

Question Concept: What are the "bite-sized" pieces of data called at the transport layer when using TCP?

Answer: Segments

Explanation: TCP divides data into segments for orderly transmission and reassembly, enabling reliability and error handling.

10. Protocol Mapping (FTP Example)

Question Concept: Which OSI layer does a common file transfer protocol like FTP operate on?

Answer: Application Layer (Layer 7)

Explanation: FTP is a high-level protocol that enables file transfers between applications, using underlying transport mechanisms (TCP).



11. Best Transport Protocol for Live Video

Question Concept: Which transport protocol is best for real-time video?

Answer: UDP

Explanation: UDP is low-latency and connectionless, ideal for streaming where speed matters more than reliability.

Task 3 – Encapsulation

Context

Encapsulation explains how data is wrapped with protocol-specific information as it moves through the network stack.

Why This Matters in Security

- Enables packet inspection and traffic analysis
- Explains how metadata is exposed
- Helps understand where filtering and monitoring occur

Task

Task 3 ○ Encapsulation

As the data is passed down each layer of the model, more information containing details specific to the layer in question is added on to the start of the transmission. As an example, the header added by the Network Layer would include things like the source and destination IP addresses, and the header added by the Transport Layer would include (amongst other things) information specific to the protocol being used. The data link layer also adds a piece on at the *end* of the transmission, which is used to verify that the data has not been corrupted on transmission; this also has the added bonus of increased security, as the data can't be intercepted and tampered with without breaking the trailer. This whole process is referred to as *encapsulation*; the process by which data can be sent from one computer to another.

Stage	Layer	Header	Trailer	Data
1	Application	L7 Header		Data (Data)
2	Presentation	L6 Header		Data (Data)
3	Session	L5 Header		Data (Data)
4	Transport	L4 Header		Data (Segments/Datagrams)
5	Network	L3 Header		Data (Packets)
6	Data Link	L2 Header	L2 Trailer	Data (Frames)
7	Physical			Data Stream (Bits)



Q&A

1. Data at Layer 2

Question Concept: How is data referred to at the Data Link layer during encapsulation?

Answer: Frames

Explanation: At layer 2, the Data Link layer wraps the network-layer packet with a header and trailer, forming a frame suitable for transmission over physical media.

2. Data at Layer 4 (UDP Selected)

Question Concept: What is the name of the data unit at the transport layer when using UDP?

Answer: Datagrams

Explanation: When UDP is used, the transport layer packages data into datagrams. These are connectionless, self-contained units that include port information and length, but without the reliability mechanisms of TCP.

3. Processing a Received Message

Question Concept: What process occurs on the receiving side to interpret a message correctly?

Answer: De-encapsulation

Explanation: Each layer removes its corresponding headers/trailers to extract the original message, reversing the encapsulation performed on the sender side.

4. Adding a Trailer

Question Concept: Which OSI layer adds a trailer during encapsulation?

Answer: Data Link Layer (Layer 2)

Explanation: The trailer is used for error detection, commonly via a CRC (Cyclic Redundancy Check), to ensure the integrity of each transmitted frame.

5. Security Aspect of Encapsulation



Question Concept: Does encapsulation provide an extra layer of security?

Answer: Yes (Aye)

Explanation: While not a replacement for encryption, encapsulation separates each protocol layer, which can help limit exposure and provide structural security, making it harder for attackers to interpret raw data without processing each layer.

Task 4 – TCP/IP Model

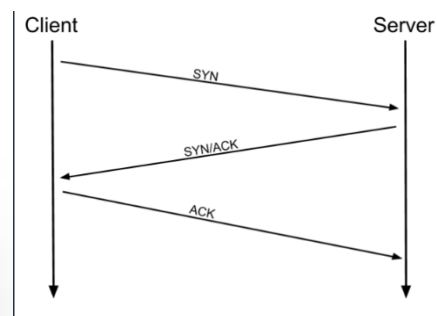
Context

This task focuses on the TCP/IP model as the real-world implementation of networking principles.

Security Insight

Understanding this model helps bridge:

- Theory → Real protocols
- Concept → Practical assessment
- Learning → Professional usage



Task

Task 4 The TCP/IP Model

The TCP/IP model is, in many ways, very similar to the OSI model. It's a few years older, and serves as the basis for real-world networking. The TCP/IP model consists of four layers: Application, Transport, Internet and Network Interface. Between them, these cover the same range of functions as the seven layers of the OSI Model.

TCP/IP
Application
Transport
Internet
Network Interface

Note: Some recent sources split the TCP/IP model into five layers -- breaking the Network Interface layer into Data Link and Physical layers (as with the OSI model). This is accepted and well-known; however, it is not officially defined (unlike the original four layers which are defined in RFC1122). It's up to you which version you use -- both are generally considered valid.

You would be justified in asking why we bother with the OSI model if it's not actually used for anything in the real-world. The answer to that question is quite simply that the OSI model (due to being less condensed and more rigid than the TCP/IP model) tends to be easier for learning the initial theory of networking.

The two models match up something like this:

OSI	TCP/IP
Application	Application
Presentation	
Session	
Transport	Transport
Network	Internet
Data Link	Network Interface
Physical	

The processes of encapsulation and de-encapsulation work in exactly the same way with the TCP/IP model as they do with the OSI model. At each layer of the TCP/IP model a header is added during encapsulation, and removed during de-encapsulation.



Q&A

1. Model Introduction Order

Question Concept: Which networking model appeared first historically?

Answer: TCP/IP

Explanation: The TCP/IP model was developed before the OSI model and became the foundation of real-world internet communication, while OSI serves as a conceptual framework.

2. TCP/IP Layer Corresponding to OSI Transport Layer

Question Concept: Which TCP/IP layer handles transport-level responsibilities?

Answer: Transport Layer

Explanation: This layer ensures end-to-end communication, providing reliable or fast delivery via protocols like TCP or UDP.

3. TCP/IP Layer Covering OSI Session Layer

Question Concept: Which TCP/IP layer incorporates session management functions?

Answer: Application Layer

Explanation: The TCP/IP Application layer combines OSI Application, Presentation, and Session functions, managing sessions and data formatting for applications.

4. Network Interface Layer Mapping

Question Concept: The TCP/IP Network Interface layer covers two OSI layers. One is Data Link; the other?

Answer: Physical Layer

Explanation: The Network Interface layer handles physical transmission (bits on media) and framing/error detection, encompassing both Data Link and Physical layers.

5. TCP/IP Layer for OSI Network Layer Functions



Question Concept: Which TCP/IP layer handles routing and logical addressing?

Answer: Internet Layer

Explanation: The Internet layer is responsible for packet addressing and routing, equivalent to the OSI Network layer.

6. TCP Protocol Type

Question Concept: What type of protocol is TCP?

Answer: Connection-based (reliable)

Explanation: TCP establishes a session and guarantees delivery, ordering, and error checking, unlike UDP.

7. Meaning of SYN

Question Concept: In TCP handshakes, what does SYN stand for?

Answer: Synchronize

Explanation: SYN signals the initiation of a connection and synchronizes sequence numbers between hosts.

8. Second Step of the Three-Way Handshake

Question Concept: During TCP connection setup, what occurs in the second step?

Answer: SYN/ACK

Explanation: The server acknowledges the SYN request from the client while sending its own synchronization request.

9. Short Name for Acknowledgment Segment

Question Concept: What is the short name for the acknowledgment in TCP handshake?

Answer: ACK

Explanation: ACK confirms receipt of data or connection requests, completing part of the handshake process.



Task 5 – Networking Tool: Ping

Context

Ping is introduced as a basic but critical network diagnostic tool.

How Pentesters Use It

- Validate host reachability
- Measure latency and packet behavior
- Establish a baseline before deeper analysis

TASK

Task 5 Networking Tools Ping

At this stage, hopefully all of the theory has made sense and you now understand the basic models behind computer networking. For the rest of the room we're going to be taking a look at some of the command line networking tools that we can use in practical applications. Many of these tools do work on other operating systems, but for the sake of simplicity, I'm going to assume that you're running Linux for the rest of this room. The first tool that we're going to look at will be the `ping` command.

The ping command is used when we want to test whether a connection to a remote resource is possible. Usually this will be a website on the internet, but it could also be for a computer on your home network if you want to check if it's configured correctly. Ping works using the ICMP protocol, which is one of the slightly less well-known TCP/IP protocols that were mentioned earlier. The ICMP protocol works on the Network layer of the OSI Model, and thus the Internet layer of the TCP/IP model. The basic syntax for ping is `ping <target>`. In this example we are using ping to test whether a network connection to Google is possible:

```
~$ ping google.com
PING google.com (216.58.198.174) 56(84) bytes of data.
```

Notice that the ping command actually returned the IP address for the Google server that it connected to, rather than the URL that was requested. This is a handy secondary application for ping, as it can be used to determine the IP address of the server hosting a website. One of the big advantages of ping is that it's pretty much ubiquitous to any network enabled device. All operating systems support it out of the box, and even most embedded devices can use ping!

Have a go at the following questions. Any questions about syntax can be answered using the man page for ping (`man ping` on Linux).

Q&A

1. Command to Ping a Website

Question Concept: How do you check if a website is reachable using ICMP requests?

Answer: `ping [domain]` (e.g., `ping bbc.co.uk`)

Explanation: The `ping` command tests connectivity to a host by sending ICMP echo requests and measuring the round-trip time of responses. This is a fundamental reconnaissance step to confirm a host is online.



```
PING muirlandoracle.co.uk (217.160.0.152) 56(84) bytes of data.  
64 bytes from 217-160-0-152.elastic-ssl.ui-r.com (217.160.0.152): icmp_seq=1 ttl=53 time=66.0 ms  
64 bytes from 217-160-0-152.elastic-ssl.ui-r.com (217.160.0.152): icmp_seq=2 ttl=53 time=68.6 ms  
64 bytes from 217-160-0-152.elastic-ssl.ui-r.com (217.160.0.152): icmp_seq=3 ttl=53 time=68.2 ms  
64 bytes from 217-160-0-152.elastic-ssl.ui-r.com (217.160.0.152): icmp_seq=4 ttl=53 time=69.3 ms  
64 bytes from 217-160-0-152.elastic-ssl.ui-r.com (217.160.0.152): icmp_seq=5 ttl=53 time=81.6 ms  
64 bytes from 217-160-0-152.elastic-ssl.ui-r.com (217.160.0.152): icmp_seq=6 ttl=53 time=68.3 ms  
64 bytes from 217-160-0-152.elastic-ssl.ui-r.com (217.160.0.152): icmp_seq=7 ttl=53 time=69.3 ms  
64 bytes from 217-160-0-152.elastic-ssl.ui-r.com (217.160.0.152): icmp_seq=8 ttl=53 time=69.2 ms
```

2. Resolving IPv4 Address

Question Concept: What is the IPv4 address associated with a domain?

Answer: 217.160.0.152 (example)

Explanation: Domain names are resolved to IP addresses using DNS. Knowing the IP helps map network infrastructure and perform further testing.

3. Changing Ping Interval

Question Concept: Which option lets you modify the time interval between sent ping requests?

Answer: -i

Explanation: The interval switch controls how frequently ICMP packets are sent, allowing you to adjust testing pace and avoid flooding the target.

4. Restricting Requests to IPv4

Question Concept: Which option forces the ping command to use IPv4 only?

Answer: -4

Explanation: Some systems have dual-stack networking (IPv4 & IPv6). Using -4 ensures packets are sent over IPv4.

5. Verbose Output

Question Concept: Which option provides more detailed ping information?

Answer: -v

Explanation: Verbose mode outputs detailed information about the ICMP process, useful for troubleshooting network connectivity.



Task 6 – Networking Tool: Traceroute

Context

Traceroute provides visibility into the path traffic takes across a network.

Security Application

- Understand routing behavior
- Identify intermediate network hops
- Gain insight into network segmentation

TASK

Task 6 **Networking Tools** Traceroute

The logical follow-up to the ping command is `traceroute`. Traceroute can be used to map the path your request takes as it heads to the target machine.

The internet is made up of many, many different servers and end-points, all networked up to each other. This means that, in order to get to the content you actually want, you first need to go through a bunch of other servers. Traceroute allows you to see each of these connections -- it allows you to see every intermediate step between your computer and the resource that you requested. The basic syntax for traceroute on Linux is this: `traceroute <destination>`

By default, the Windows traceroute utility (`tracert`) operates using the same ICMP protocol that ping utilises, and the Unix equivalent operates over UDP. This can be altered with switches in both instances.

```
~$ traceroute google.com
traceroute to google.com (216.58.205.46), 30 hops max, 60 byte packets
 1 _gateway (172.16.255.254)  14.883 ms  15.401 ms  15.551 ms
 2 193.60.160.253 (193.60.160.253)  1.464 ms  1.872 ms  2.026 ms
 3 193.60.168.92 (193.60.168.92)  3.084 ms  4.093 ms  4.814 ms
 4 ge-0-3-2.dund-ban1.ja.net (146.97.128.85)  4.768 ms  4.253 ms  4.715 ms
 5 ae1.dund-ban3.ja.net (146.97.64.97)  10.320 ms  5.114 ms  10.589 ms
 6 ae24.leedaq-sbr2.ja.net (146.97.37.181)  11.160 ms  10.855 ms  10.766 ms
 7 ae29.lowdss-sbr1.ja.net (146.97.33.50)  11.992 ms  11.048 ms  10.746 ms
 8 ae31.londtw-sbr2.ja.net (146.97.33.30)  13.558 ms  13.245 ms  13.561 ms
 9 ae28.londtt-sbr1.ja.net (146.97.33.61)  13.541 ms  13.229 ms  11.410 ms
10 72.14.205.74 (72.14.205.74)  15.143 ms  14.607 ms  13.865 ms
11 74.125.242.97 (74.125.242.97)  13.263 ms  74.125.242.65 (74.125.242.65)  12.553 ms  12.904 ms
12 172.253.71.191 (172.253.71.191)  13.943 ms  12.833 ms  172.253.71.189 (172.253.71.189)  12.631 ms
13 lhr48s23-in-f14.1e100.net (216.58.205.46)  13.227 ms  12.258 ms  12.482 ms
```

You can see that it took 13 hops to get from my router (`_gateway`) to the Google server at 216.58.205.46

Now it's your turn. As with before, all questions about switches can be answered with the man page for traceroute (`man traceroute`).

Q&A

1. Observing the Request Path

Question Concept: Can you see the network path your request takes?

Answer: N/A (No answer needed)

Explanation: Traceroute shows each hop a packet takes toward the destination. While in



practice the path may vary or some hops may be hidden, the exercise is meant to illustrate how network paths are mapped.

2. Specifying a Network Interface

Question Concept: Which option allows you to specify a network interface when running traceroute?

Answer: -i

Explanation: On systems with multiple interfaces, this switch directs traceroute to use a specific interface, enabling controlled testing from a chosen network source.

3. Using TCP SYN Requests

Question Concept: Which switch allows traceroute to use TCP SYN packets instead of ICMP?

Answer: -T

Explanation: Using TCP SYN requests can bypass firewalls that block ICMP, allowing the route to be traced through firewalled networks while mimicking standard connection attempts.

4. Default Layer of Operation

Question Concept: Which TCP/IP layer does traceroute operate on by default in Windows?

Answer: Internet Layer

Explanation: Traceroute relies on IP packet headers to determine the path across networks. It measures TTL (Time To Live) expiration to identify hops, which corresponds to the Internet layer's routing functionality.

Task 7 – Networking Tool: WHOIS

Context

WHOIS queries provide registration and ownership information about domains and IP ranges.



Why This Is Useful

- Supports OSINT and reconnaissance
- Helps identify organizational boundaries
- Assists in scope validation

TASK

Task 7 Networking Tools WHOIS

Domain Names -- the unsung saviours of the internet.

Can you imagine how it would feel to remember the IP address of every website you want to visit? Horrible thought.

Fortunately, we've got domains.

We'll talk a little bit more about how this works in the next task, but for now suffice to know that a domain translates into an IP address so that we don't need to remember it (e.g. you can type tryhackme.com, rather than the TryHackMe IP address). Domains are leased out by companies called Domain Registrars. If you want a domain, you go and register with a registrar, then lease the domain for a certain length of time.

Enter Whois.

Whois essentially allows you to query who a domain name is registered to. In Europe personal details are redacted; however, elsewhere you can potentially get a great deal of information from a whois search.

For Free Users using the AttackBox, there is a [web version](#) of the whois tool.

(Note: You may need to install whois before using it. On Debian based systems this can be done with `sudo apt update && sudo apt-get install whois`)

Whois lookups are very easy to perform. Just use `whois <domain>` to get a list of available information about the domain registration:

```
- $ whois bbc.co.uk

Domain name:
  bbc.co.uk

Data validation:
  Nominet was able to match the registrant's name and address against a 3rd party data source on 12-Jun-2014

Registrar:
  British Broadcasting Corporation [Tag = BBC]
  URL: http://www.bbc.co.uk

Relevant dates:
  Registered on: before Aug-1996
  Expiry date:  13-Dec-2025
  Last updated: 29-Oct-2016

Registration status:
  Registered until expiry date.

Name servers:
  ns3.bbc.co.uk          156.154.66.17  2610:a1:1015::17
  ns3.bbc.net.uk         156.154.67.17  2001:502:4612::17
  ns4.bbc.co.uk          156.154.67.17  2001:502:4612::17
  ns4.bbc.net.uk         156.154.67.17  2001:502:4612::17

WHOIS lookup made at 02:22:04 07-Mar-2020
```

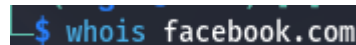


Q&A

1. Performing a WHOIS Lookup

Question Concept:How do you retrieve domain registration details?

Answer:N/A (No answer needed)



Explanation:A WHOIS lookup is used to retrieve publicly available domain registration information, including the registrant, registrar, registration dates, and administrative contacts.

2. Registrant Postal Code

Question Concept:What is the postal code of a domain registrant?

Answer:94025 (facebook.com)



Explanation:Postal codes indicate the geographical area of the domain owner. This information is useful in OSINT and reconnaissance to understand the physical location of the entity behind the domain.

3. Domain Registration Date

Question Concept:When was a domain first registered?

Answer:29/03/1997 (facebook.com)



Explanation:The domain registration date provides historical insight into the domain's age and lifecycle, which can help assess trustworthiness and identify suspicious ownership changes.

4. Registrant City



Question Concept:What city is a domain registrant located in?

Answer:Redmond (microsoft.com)

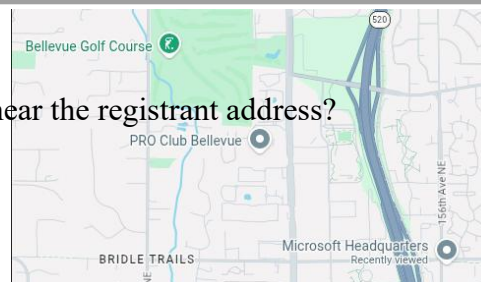


Explanation:Identifying the registrant's city helps security analysts validate organizational information and map infrastructure during reconnaissance activities.

5. Nearby Landmarks (OSINT Insight)

Question Concept:What notable landmark is located near the registrant address?

Answer:Bellevue Golf Course (microsoft.com)



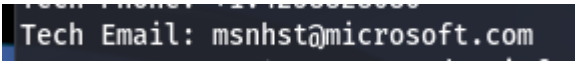


Explanation: OSINT techniques often include correlating domain data with physical landmarks to enhance situational awareness and support intelligence gathering.

6. Technical Contact Email

Question Concept: What is the registered technical contact email for a domain?

Answer: msnhst@microsoft.com



Explanation: Technical contact emails are part of standard WHOIS records and are used for administrative and technical communication related to domain management.

Task 8 – Networking Tool: Dig

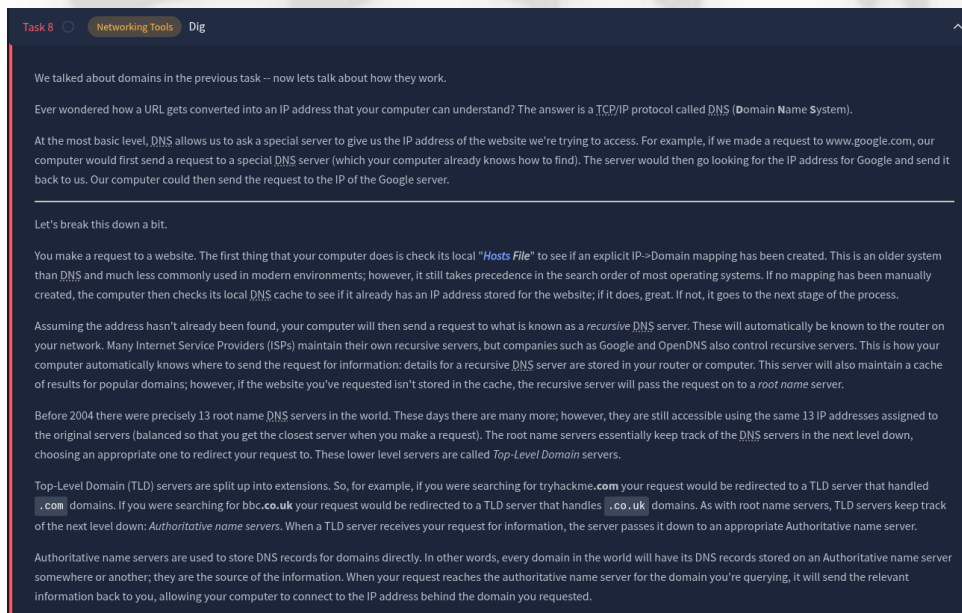
Context

Dig is used to query DNS records and analyze name resolution behavior.

Security Relevance

- Understand DNS infrastructure
- Identify misconfigurations
- Support passive reconnaissance

TASK



Task 8 Networking Tools Dig

We talked about domains in the previous task – now lets talk about how they work.

Ever wondered how a URL gets converted into an IP address that your computer can understand? The answer is a TCP/IP protocol called **DNS (Domain Name System)**.

At the most basic level, DNS allows us to ask a special server to give us the IP address of the website we're trying to access. For example, if we made a request to `www.google.com`, our computer would first send a request to a special **DNS** server (which your computer already knows how to find). The server would then go looking for the IP address for Google and send it back to us. Our computer could then send the request to the IP of the Google server.

Let's break this down a bit.

You make a request to a website. The first thing that your computer does is check its local **"Hosts File"** to see if an explicit IP->Domain mapping has been created. This is an older system than DNS and much less commonly used in modern environments; however, it still takes precedence in the search order of most operating systems. If no mapping has been manually created, the computer then checks its local DNS cache to see if it already has an IP address stored for the website; if it does, great. If not, it goes to the next stage of the process.

Assuming the address hasn't already been found, your computer will then send a request to what is known as a **recursive DNS** server. These will automatically be known to the router on your network. Many Internet Service Providers (ISPs) maintain their own recursive servers, but companies such as Google and OpenDNS also control recursive servers. This is how your computer automatically knows where to send the request for information: details for a recursive DNS server are stored in your router or computer. This server will also maintain a cache of results for popular domains; however, if the website you've requested isn't stored in the cache, the recursive server will pass the request on to a **root name** server.

Before 2004 there were precisely 13 root name DNS servers in the world. These days there are many more; however, they are still accessible using the same 13 IP addresses assigned to the original servers (balanced so that you get the closest server when you make a request). The root name servers essentially keep track of the DNS servers in the next level down, choosing an appropriate one to redirect your request to. These lower level servers are called **Top-Level Domain** servers.

Top-Level Domain (TLD) servers are split up into extensions. So, for example, if you were searching for `tryhackme.com` your request would be redirected to a TLD server that handled `.com` domains. If you were searching for `bbc.co.uk` your request would be redirected to a TLD server that handles `.co.uk` domains. As with root name servers, TLD servers keep track of the next level down: **Authoritative name** servers. When a TLD server receives your request for information, the server passes it down to an appropriate Authoritative name server.

Authoritative name servers are used to store DNS records for domains directly. In other words, every domain in the world will have its DNS records stored on an Authoritative name server somewhere or another; they are the source of the information. When your request reaches the authoritative name server for the domain you're querying, it will send the relevant information back to you, allowing your computer to connect to the IP address behind the domain you requested.



Q&A – Domain Name System (DNS)

1. DNS Definition

Question Concept: What does DNS stand for?

Answer: Domain Name System

Explanation: DNS is a fundamental networking service that translates human-readable domain names into IP addresses, allowing devices to locate and communicate with servers on a network or the internet

2. First DNS Server Queried

Question Concept: What is the first type of DNS server a computer queries when resolving a domain?

Answer: Recursive DNS Server

Explanation: A recursive DNS server acts on behalf of the client by querying other DNS servers until it finds the requested IP address or determines that the domain does not exist.

3. DNS Server for Domain Extensions

Question Concept: Which type of DNS server contains records for domain extensions such as .com or .co.uk?

Answer: Top-Level Domain (TLD) Server

Explanation: TLD servers store information related to specific domain extensions and direct queries to the appropriate authoritative DNS servers.

4. First Local Lookup Location

Question Concept: Where does a computer first check for a domain's IP address before querying DNS servers?

Answer: Hosts File

Explanation: The hosts file is a local file that maps domain names to IP addresses. If an entry exists, the system uses it without performing external DNS queries.



5. Public DNS Servers (Research Insight)

Question Concept: What is the second public DNS server IP operated by Google?

Answer: 8.8.4.4

Explanation: Google provides public DNS services for faster and more reliable name resolution, commonly used for testing, troubleshooting, and improving DNS performance.

Google Public DNS is a free, fast, and secure recursive DNS resolver offered by Google to enhance internet browsing speed, security, and accuracy worldwide. By using Anycast routing, it directs traffic to the nearest data center, often reducing latency. 

Primary IP Addresses (IPv4):

- 8.8.8.8
- 8.8.4.4

6. DNS Time To Live (TTL)

Question Concept: What value is shown by a DNS query for a TTL of 24 hours?

Answer: 86400

Explanation: TTL is measured in seconds and determines how long a DNS record is cached. A value of 86400 seconds represents 24 hours.

Task 9 Further Reading

Task 9 ☐ Further Reading

That's us completed our whirlwind tour of networking basics. Hopefully you've found it informative!

Networking is one of those things that you just need to learn. We've covered the very basics, but it would be a very good idea to continue to learn by yourself.

In terms of further information, feel free to reach out in the TryHackMe Discord if you want any help with the contents of this room. Additionally, if you want to expand your knowledge of networking theory, the [CISCO Self Study Guide by Steve McQuery](#) is a great resource to work from. There may be a more up to date version available; however, this edition is cheap, readily available, and most importantly, still very relevant. Whilst it is designed to as a study guide for the CCNA exam, that book serves equally well as a very rounded introduction to networking principles.

Answer the questions below

Read the final thoughts

Check

Methodology & Technical Thinking

Rather than solving tasks mechanically, this room encourages:

- Observing network behavior
- Understanding protocol roles



- Interpreting output instead of memorizing commands

This mindset aligns directly with professional penetration testing workflows.

Key Takeaways

- Networking fundamentals are non-negotiable in cybersecurity
 - Models are thinking frameworks, not memorization exercises
 - Simple tools provide powerful insight when used correctly
 - Ethical documentation adds long-term career value
-

Real-World Application

These concepts are applied daily in:

- Network reconnaissance
- Incident response
- Security assessments
- Technical reporting and communication

Strong fundamentals improve both offensive and defensive security work.

Final Notes

This documentation is intentionally high-level and ethical.

It demonstrates learning outcomes and professional reasoning without exposing lab content, answers, or sensitive details.